



Application of deep learning based multi-fidelity surrogate model to robust aerodynamic design optimization



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ARTICLE INFO

Article history:

Received 18 March 2019

Received in revised form 10 June 2019

Accepted 2 July 2019

Available online 4 July 2019

Keywords:

Multi-fidelity surrogate model

Deep learning

Deep belief network

Improved PSO algorithm

Aerodynamic design optimization

ABSTRACT

In the present work, a multi-fidelity surrogate-based optimization framework is proposed, and then applied to the robust optimizations for airfoil and wing under uncertainty of Mach number. DBN (deep belief network) is employed as the low-fidelity model, and the k -step contrastive divergence algorithm is used for training the network. By virtue of the well trained DBN model and high-fidelity data, a linear regression multi-fidelity surrogate model is established. Verification results indicate that the multi-fidelity surrogate model obtains more accurate predictions than the DBN model and is highly reliable as a prediction model. The multi-fidelity surrogate model is embedded into an improved PSO (particle swarm optimization) algorithm framework, and is updated in each iteration of the robust optimization processes for both airfoil and wing. Comparisons between multi-fidelity surrogate predictions and CFD results indicate that, the multi-fidelity surrogate predictions tend to approach the CFD results as the iteration number increases. The robust optimization results of airfoil and wing demonstrate that, the multi-fidelity surrogate model performs very well as a prediction model, and improves the optimization efficiency obviously.

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1. Introduction

Aerodynamic design optimization of commercial aircraft is a process with a large number of iterations. Usually, CFD methods are employed in each iteration to evaluate objective functions to guide the optimization directions. However, highly accurate CFD methods such as DES (detached eddy simulation) method and LES (large eddy simulation) method are very expensive and time-consuming, and thus can hardly applied to aerodynamic design optimization of commercial aircraft. Rapid analytical methods such as panel method and full potential method can obtain the results in a short time, but the accuracy of these methods is relatively low and the applicability is limited. Although RANS (Reynolds-averaged Navier-Stokes) method provides more accurate results than those rapid analytical methods and costs less computational time as well as resources than those highly accurate methods, the computational costs of RANS method are still enormous for aerodynamic design optimization of commercial aircraft. In order to reduce computational costs and improve the optimization efficiency, surrogate models have been studied and ap-

plied in aerodynamic design optimization more and more in recent years.

Surrogate model is a type of approximation model, which is developed to substitute for those complicated and time-consuming analyses and computations during the optimization process. Optimizations employing surrogate models are called SBOs (surrogate-based optimizations). Over the years, a number of surrogate models have been proposed and applied in SBOs, such as RSM (response surface model) [1], Kriging [2], RBFs (radial basis functions) [3], ANN (artificial neural network) [4], PCE (polynomial chaos expansion) [5], SVR (support vector regression) [6] etc. These surrogate models have been introduced into the field of aerodynamic design optimization. Pehlivanoglu et al. [7] conducted aerodynamic design optimizations for airfoil, multi-element airfoil and active flow control of airfoil using RSM surrogate model, and their results showed reductions of 50%, 70% and 50% respectively in computational amount for the three optimization cases. Zhou et al. [8] accomplished the maximum lift-drag-ratio optimization for airfoil via RBFs surrogate model, and their results indicated obvious reduction in computational amount. Boopathy et al. [9] combined Kriging model and PCE model for airfoil optimization, and their results indicated that the surrogate model improved the optimization efficiency dramatically. Han et al. [10,11] conducted optimizations for active flow control of airfoil and aerodynamic-structural design

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of subsonic wing using RSM, Kriging and RBFs surrogate models, and obtained satisfying results on the premise of reducing computational costs.

The result of SBO is highly dependent on the approximation accuracy of the surrogate model. In order to improve the accuracy of the surrogate model and reduce high-fidelity computations, various multi-fidelity surrogate models have been proposed by combining data from different fidelities. Kennedy and O'Hagan [12] proposed a Bayesian approach in terms of Gaussian processes in order to improve the efficiency of prediction by combining computer codes at different levels, and the proposed KOH method is applied to an example using two versions of an oil reservoir simulator. Zimmermann et al. [13] proposed a simplified method for multi-fidelity surrogate CoKriging models to reduce the number of total model tuning parameters. Their predictors were compared to KOH method, and performed in all given examples comparable or even better. J. Toal et al. [14] conducted multipoint optimization for airfoil using the CoKriging multi-fidelity surrogate model, and their results showed a computational amount reduction of 60% on the premise of ensuring the prediction accuracy. Han et al. [15] proposed an improved CoKriging multi-fidelity surrogate and applied it to aerodynamic performance prediction of RAE 2822 airfoil. The comparison results indicated that the improved CoKriging surrogate model obtained more accurate prediction results than Kriging surrogate model. Han et al. [16] proposed a hierarchical Kriging model and used it for modeling the aerodynamic data of RAE 2822 airfoil and an industrial transport aircraft configuration. The numerical examples show that it is efficient, accurate and robust. Courrier et al. [17] tested five multi-fidelity models, and compared the performances of these five models in solving three mechanical examples. They concluded that the additive-bridge function method, Zimmermann's CoKriging method and the hierarchical Kriging method are efficient for creating a surrogate model. Han et al. [18] proposed a novel variable-fidelity surrogate model of which the key ingredient is the theory and algorithm of a multi-level hierarchical Kriging model, the multi-fidelity surrogate model incorporates simulation data from levels of fidelity on multilevel computational grids. Their proposed multi-fidelity surrogate model was applied to aerodynamic shape optimization of a NACA0012 airfoil and an ONERA M6 wing in transonic flows, and the results indicated that the proposed multi-fidelity surrogate model can significantly improve the optimization efficiency and apparently outperforms the existing single-fidelity or two-level-fidelity method. Aside from these multi-fidelity surrogate models, other multi-fidelity surrogate models such as ARP (adaptive response prediction) [19], NIPC (non-intrusive polynomial chaos) [20] have also been proposed and developed. Although efforts have been made to develop the multi-fidelity surrogate model, there is still space for investigating other multi-fidelity surrogate models to expand the arsenal of multi-fidelity surrogate model.

Given the above, this study aims to develop an efficient multi-fidelity surrogate model distinguished from the conventional multi-fidelity surrogate models which are built based on Kriging or RBFs etc. In this study, a DBN (deep belief network)-based multi-fidelity surrogate model is proposed, and applied to the robust aerodynamic design optimizations of airfoil and wing under uncertainty of Mach number. DBN is employed as the low-fidelity model, and then a linear regression multi-fidelity surrogate model is built by combining the well trained DBN model and high-fidelity data. The multi-fidelity surrogate model is verified through the McCormick function, the dataset from fluidized-bed process, and predictions of lift-to-drag ratios for a set of airfoils. By employing the multi-fidelity surrogate model to evaluate fitness functions and updating the multi-fidelity surrogate model in each iteration, an improved PSO (particle swarm optimization) algorithm

framework is proposed for the robust optimizations of airfoil and wing.

This paper is organized as follows. In Section 2, the establishment of the multi-fidelity surrogate model is introduced. In Section 3, numerical method and optimization algorithm are introduced and validated. In Section 4, verification of the multi-fidelity surrogate model is presented. In Section 5, the robust optimizations for airfoil and wing are performed via the improved PSO framework incorporating the multi-fidelity surrogate model. Finally, in Section 6, some conclusions are drawn according to the previous results and analyses.

2. Multi-fidelity surrogate model

In this section, the flowchart of the multi-fidelity surrogate-based optimization is introduced, and the construction of the multi-fidelity surrogate model as well as the establishment of the DBN model used for airfoil and wing optimizations are presented.

2.1. Architecture of the multi-fidelity surrogate-based optimization

Fig. 1 shows the architecture of optimization using the deep learning based multi-fidelity surrogate model. The optimization is implemented following steps as below.

- (1) Generate a number of samples as the original samples ($\mathbf{X}$), and then analyze or compute the original samples via high-fidelity model ($f_H(\mathbf{X})$, e.g. CFD simulations for aerodynamic design optimization);
- (2) Divide the original samples into a majority part (e.g. 80%) and a remaining minority part (e.g. 20%), take the minority part ($\mathbf{X}_H, f_H(\mathbf{X}_H)$) as high-fidelity data and use the majority part ($\mathbf{X}_L, f_H(\mathbf{X}_L)$) to build a preliminary database;
- (3) Build a DBN and train the network using the preliminary database as training samples, then take the well trained network as the low-fidelity model;
- (4) Construct the multi-fidelity surrogate model on the basis of the high-fidelity data and the low-fidelity model;
- (5) Embed the multi-fidelity surrogate model into an optimizer (e.g. GA, PSO) to evaluate the objective functions (or fitness functions), and update the population or particle swarm;
- (6) Judge whether the optimization is converged, if yes, end the optimization, otherwise go to step (7);
- (7) Select part (e.g. 20%) of the population/particle swarm as additional samples, and analyze or compute them via high-fidelity model;
- (8) Take half results of the additional samples as high-fidelity data, and update the database by adding in the other half of the additional samples and results;
- (9) Update the low-fidelity model by retraining the DBN on the basis of the updated database, and then reconstruct the multi-fidelity surrogate model using the updated high-fidelity data and the updated low-fidelity model;
- (10) Go to step (5) and continue the optimization.

Infill sampling criterion and constraint handling are important issues for SBOs (surrogate-based optimizations). Forrester et al. [21] summarized several popular infill sampling criteria, and provided some examples of applying these criteria to SBOs. Sasena et al. [22] reviewed a few different infill sampling criteria, and investigated the effect of them on the optimization results. Liu et al. [23] proposed a parallel infilling strategy which uses a combination of multiple infill criteria with each criterion choosing a different sample point, and applied it to SBOs of transonic wings. Their results indicated that the proposed method is more effective than

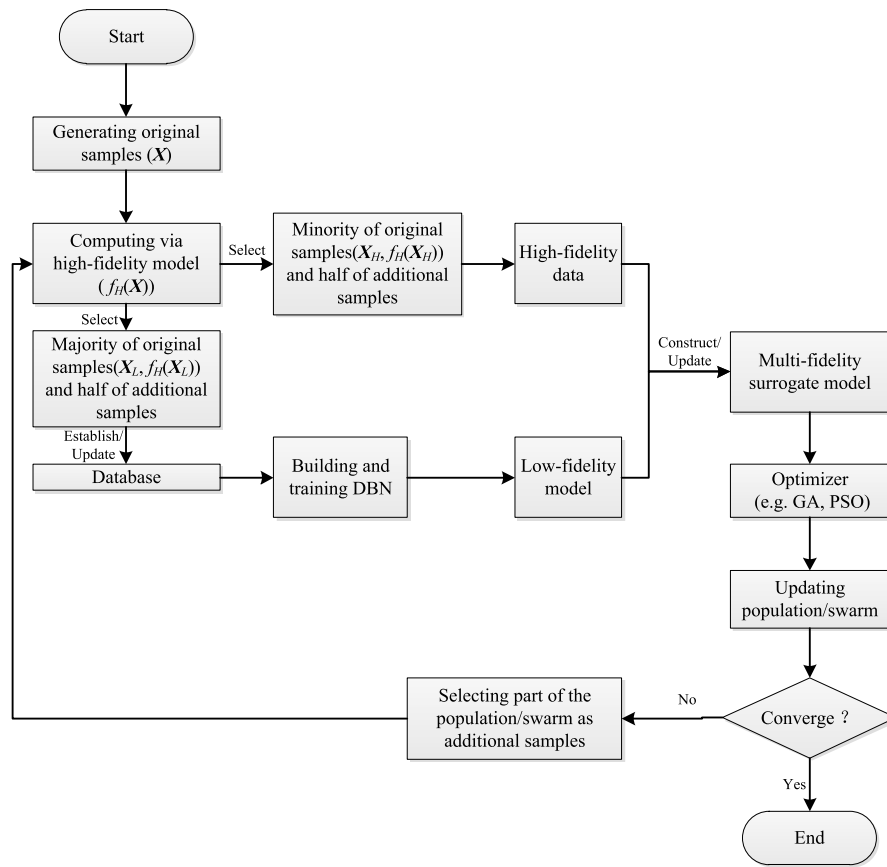


Fig. 1. Architecture of optimization using the deep learning based multi-fidelity surrogate model.

the existing parallel infilling strategies for SBOs of transonic wings. A traditional method for constraint handling is to transform the constrained problem into an unconstrained one by penalizing the constraints with a penalty function. The penalty is included if any one of the constraints predicts a violation for limiting the objective improvement only in the strictly feasible space. For simply constrained problems, penalty methods can perform well; however, for deceptive constraints or poorly modeled constraint functions, penalty methods can be misled or wrongly applied. Parr et al. [24, 25] investigated a probabilistic approach to constraint handling, and their results indicated that the probabilistic method outperformed the penalty method on a number of problems. Zhang et al. [26] investigated three different constrained aggregation approaches for wing structural design, and combined them with an in-house SBO code within the framework of a gradient-free optimization. They concluded from their test cases that the most efficient way of handling large-scale constraints for wing SBO is to aggregate constraints within sub-domains and with a relatively large constant parameter.

In this study, the infilling samples are selected from the particles in the current iteration. In each iteration, 20% of the particles are selected by adopting the multiple-point infilling criteria, which selects 10% particles near the current optimal particles and 10% particles with the largest prediction variances. A type of penalty method [27] is employed for constraint handling. By implementing the multi-fidelity surrogate-based optimization, in each iteration, new samples are added to update the DBN model namely the low-fidelity model, and new high-fidelity data are obtained simultaneously to update the multi-fidelity surrogate model. Therefore, the predictions are expected to become more accurate as the iteration number increases.

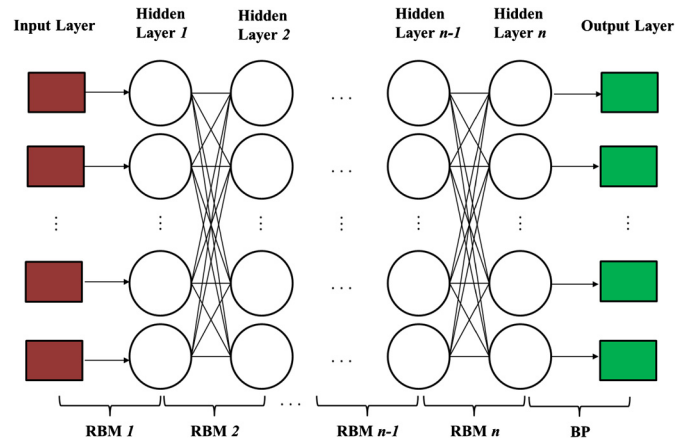


Fig. 2. Structure of the DBN.

2.2. Deep learning model

In recent years, DNNs (deep neural networks) [28–30] have become an important branch of ANNs, and have been studied by more and more scholars. Different from conventional ANNs, DNNs employ much more layers and thus have deeper net structures. By introducing particular mathematical operations and activation functions, DNNs have shown great potential in extracting information from the input data and fitting nonlinear functions.

In this study, a type of DNN called DBN [31] is introduced for building the low-fidelity model. Fig. 2 shows the structure of the DBN, it consists of one input layer, n hidden layers and one out-

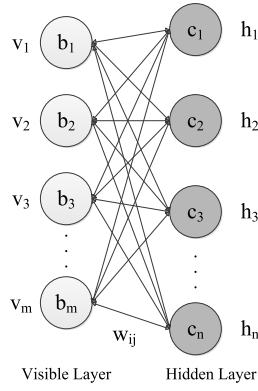


Fig. 3. Structure of a single RBM.

put layer. The DBN is trained through a greedy learning algorithm, the training process can be divided into two steps. Firstly, a series of RBMs (restricted Boltzmann machines) are trained layer by layer forward through unsupervised learning to preliminarily construct the DBN, this procedure is also called pre-training of DBN. Secondly, a BP (back propagation) neural network is established between the last two layers of the DBN to tune weights of the entire network backward in a supervised manner, this procedure is also called fine tuning of DBN.

Fig. 3 shows the structure of a single RBM, it is composed of a visible layer and a hidden layer. The two layers are connected bidirectionally, while unit in each single layer is not connected with other units in the same layer. W_{ij} ($i = 1, 2, \dots, m$; $j = 1, 2, \dots, n$) is the weight associated with the connection between the hidden unit h_j and the visible unit v_i , b_i is the bias weight for the visible unit v_i , and c_j is the bias weight for the hidden unit h_j .

Given these, the energy function of a RBM $(\mathbf{v}, \mathbf{h})$ is defined as:

$$E(\mathbf{v}, \mathbf{h}) = -\sum_{i=1}^m b_i v_i - \sum_{j=1}^n c_j h_j - \sum_{j=1}^n \sum_{i=1}^m W_{ij} v_i h_j \quad (1)$$

According to the energy function, joint probability density function between the visible layer and the hidden layer is obtained as:

$$p(\mathbf{v}, \mathbf{h}) = \frac{e^{-E(\mathbf{v}, \mathbf{h})}}{\sum_{\mathbf{v}} \sum_{\mathbf{h}} e^{-E(\mathbf{v}, \mathbf{h})}} \quad (2)$$

Thus, the marginal probability of the visible layer can be expressed as:

$$p(\mathbf{v}) = \frac{\sum_{\mathbf{h}} e^{-E(\mathbf{v}, \mathbf{h})}}{\sum_{\mathbf{v}} \sum_{\mathbf{h}} e^{-E(\mathbf{v}, \mathbf{h})}} \quad (3)$$

Because units in the same layer are not connected with each other, they are independent from each other. Then conditional probabilities for $\mathbf{v}$ and $\mathbf{h}$ are obtained as:

$$\begin{aligned} p(\mathbf{h}|\mathbf{v}) &= \prod_n p(h_n = 1|\mathbf{v}) \\ p(\mathbf{v}|\mathbf{h}) &= \prod_m p(v_m = 1|\mathbf{h}) \end{aligned} \quad (4)$$

When the status of the visible layer is given, the probability of activation for the hidden unit h_j can be obtained as:

$$p(h_j = 1|\mathbf{v}) = f_a \left(c_j + \sum_{i=1}^m v_i W_{ij} \right) \quad (5)$$

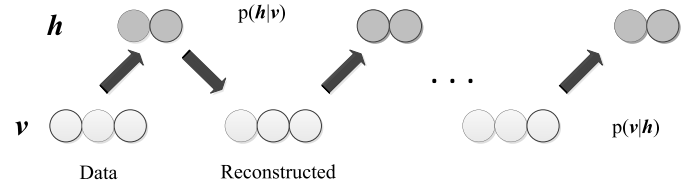


Fig. 4. Training process for a single RBM using the k -step contrastive divergence algorithm.

where f_a is the activation function. In this study, exponential linear unit described in Eq. (6) is selected as the activation function.

$$f_a(x) = \begin{cases} x & x > 0 \\ \alpha(e^x - 1) & x \leq 0 \end{cases} \quad (6)$$

Since the connection between the visible layer and the hidden layer are bidirectional, similarly, the probability of activation for the visible unit v_i can be obtained as:

$$p(v_i = 1|\mathbf{h}) = f_a \left(b_i + \sum_{j=1}^n h_j W_{ij} \right) \quad (7)$$

In order to train the RBM, the k -step contrastive divergence algorithm [32] incorporating Gibbs sampling method [33] is employed. By using this algorithm, estimations for all the weights are able to be obtained in k iterations. Fig. 4 shows the training process for a single RBM using the k -step contrastive divergence algorithm.

The training is implemented through the following steps:

- (1) Initialize the visible layer $(\mathbf{W}, \mathbf{b}, \mathbf{c})$, set k and learning rate ε , then assign original sample values to visible units marked as $\mathbf{v}^{(0)}$;
- (2) Calculate the probability of activation $p(h_j|\mathbf{v}^{(0)})$ for each hidden unit h_j ($j = 1, 2, \dots, n$) via Eq. (5);
- (3) By Gibbs sampling method, draw sample

$$h_j^{(0)} \sim p(h_j|\mathbf{v}^{(0)}) \quad j = 1, 2, \dots, n \quad (8)$$

- (4) Calculate the probability of activation $p(v_i|\mathbf{h}^{(0)})$ for each visible unit v_i ($i = 1, 2, \dots, m$) via Eq. (7), and reconstruct the visible layer;
- (5) By Gibbs sampling method, draw sample

$$v_i^{(1)} \sim p(v_i|\mathbf{h}^{(0)}) \quad i = 1, 2, \dots, m \quad (9)$$

- (6) Repeat steps (2)–(6) for $k - 1$ times;
- (7) Update all the weights as below:

$$\begin{aligned} w_{ji}^{(k)} &= w_{ji}^{(0)} + \varepsilon [p(h_j|\mathbf{v}^{(0)})v_i^{(0)} - p(h_j|\mathbf{v}^{(k)})v_i^{(k)}] \\ b_i^{(k)} &= b_i^{(0)} + \varepsilon (v_i^{(0)} - v_i^{(k)}) \quad i = 1, 2, \dots, m; \quad j = 1, 2, \dots, n \\ c_j^{(k)} &= c_j^{(0)} + \varepsilon [p(h_j|\mathbf{v}^{(0)}) - p(h_j|\mathbf{v}^{(k)})] \end{aligned} \quad (10)$$

The RBMs are trained layer by layer to obtain the initial weights and bias weights of the entire network. The original sample values are assigned to the visible layer of the first RBM, and the hidden layer of the first RBM is taken as the visible layer of the second RBM. And so on, the hidden layer of the previous RBM is taken as the visible layer of the next RBM. After obtaining initial values of the entire network through training the RBMs, the BP network is employed to tune weights of the entire network backward in a supervised manner. By employing RBMs, DBN overcomes the shortcomings that BP network traps into local optimum easily and costs a long training time caused by initializing the network randomly.

2.3. Construction of the multi-fidelity surrogate model

In this study, the original samples are divided into two parts, namely $\mathbf{X}_H = \{\mathbf{x}_H^{(1)}, \dots, \mathbf{x}_H^{(n_H)}\}$ and $\mathbf{X}_L = \{\mathbf{x}_L^{(1)}, \dots, \mathbf{x}_L^{(n_L)}\}$, where the subscripts “H” and “L” denote high-fidelity and low-fidelity, respectively. Both $\mathbf{X}_H$ and $\mathbf{X}_L$ are evaluated by the high-fidelity model, the corresponding function values are denoted as $f_H(\mathbf{X}_H)$ and $f_H(\mathbf{X}_L)$, respectively. Then $\{\mathbf{X}_L, f_H(\mathbf{X}_L)\}$ are used to build and train the DBN model. After the DBN model namely the low-fidelity model is well trained, it is applied to evaluate the high-fidelity samples, and the corresponding function values are denoted as $f_L(\mathbf{X}_H)$.

Given the above, a type of linear regression multi-fidelity surrogate model [34] is built, and it can be described in the form as follows:

$$\hat{f}_H(\mathbf{x}) = \rho f_L(\mathbf{x}) + \hat{\delta}(\mathbf{x}) \quad (11)$$

where ρ is the scale factor for the low-fidelity model, and $\hat{\delta}(\mathbf{x})$ is the discrepancy function. By using the well trained DBN model as the low-fidelity model, high-fidelity samples $\mathbf{X}_H$ and other prediction points are able to be evaluated by the low-fidelity model, and thus it is unnecessary for the low-fidelity model to have an explicit expression.

The discrepancy function is expressed in the form of polynomials as below:

$$\hat{\delta}(\mathbf{x}) = \sum_{j=1}^p \xi_j(\mathbf{x}) b_j \quad (12)$$

where $\xi_j(\mathbf{x})$ represents the j -th monomial basis, and b_j is the unknown coefficient. Since the number of high-fidelity samples is usually small, the number of terms p is correspondingly limited.

The form of the linear regression multi-fidelity surrogate model is similar to that of the hierarchical Kriging model proposed by Han et al. [16], but the trainings of these two multi-fidelity surrogate models are different, namely, the unknown coefficients and solving methods for these unknown coefficients are different. For the linear regression multi-fidelity surrogate model, the unknown coefficients are ρ and b_j ($j = 1, 2, \dots, p$), which have to be determined to obtain the complete expression of the multi-fidelity surrogate model.

According to the standard least-squares method, the errors between multi-fidelity surrogate predictions and the high-fidelity results can be expressed as:

$$\begin{aligned} e^{(i)} &= f_H(\mathbf{x}_H^{(i)}) - \hat{f}_H(\mathbf{x}_H^{(i)}) \\ &= f_H(\mathbf{x}_H^{(i)}) - \rho f_L(\mathbf{x}_H^{(i)}) - \sum_{j=1}^p \xi_j(\mathbf{x}_H^{(i)}) b_j \quad i = 1, 2, \dots, n_H \end{aligned} \quad (13)$$

Eq. (13) can also be written in the form of vector and matrix as follows:

$$\mathbf{e} = \mathbf{Y} - \mathbf{X}\mathbf{B} \quad (14)$$

Where,

$$\begin{aligned} \mathbf{e} &= \begin{bmatrix} e^{(1)} \\ \vdots \\ e^{(n_H)} \end{bmatrix}, \quad \mathbf{Y} = \begin{bmatrix} f_H(\mathbf{x}_H^{(1)}) \\ \vdots \\ f_H(\mathbf{x}_H^{(n_H)}) \end{bmatrix}, \\ \mathbf{X} &= \begin{bmatrix} f_L(\mathbf{x}_H^{(1)}) & \xi_1(\mathbf{x}_H^{(1)}) & \cdots & \xi_p(\mathbf{x}_H^{(1)}) \\ \vdots & \vdots & \ddots & \vdots \\ f_L(\mathbf{x}_H^{(n_H)}) & \xi_1(\mathbf{x}_H^{(n_H)}) & \cdots & \xi_p(\mathbf{x}_H^{(n_H)}) \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} \rho \\ b_1 \\ \vdots \\ b_p \end{bmatrix} \end{aligned} \quad (15)$$

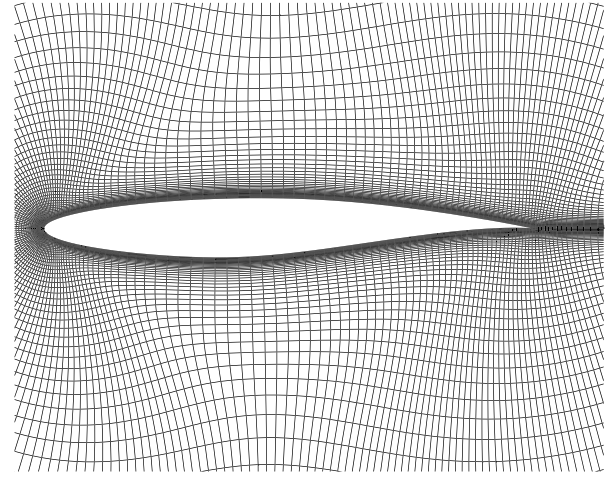


Fig. 5. Computational grid around RAE 2822 airfoil.

Here, in Eq. (15), $\mathbf{e}$ is the vector of residual errors, $\mathbf{Y}$ is the vector of high-fidelity data, $\mathbf{X}$ is the augmented design matrix, and $\mathbf{B}$ is the vector of unknown coefficients. The unknown coefficients ρ and b_j ($j = 1, 2, \dots, p$) are able to be obtained by minimizing the square sum of errors as follows:

$$\underset{\mathbf{B}}{\text{minimize}} \mathbf{e}^T \mathbf{e} = (\mathbf{Y} - \mathbf{X}\mathbf{B})^T (\mathbf{Y} - \mathbf{X}\mathbf{B}) \quad (16)$$

By implementing the standard regression technique, the unknown coefficients are obtained as:

$$\mathbf{B} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y} \quad (17)$$

The prediction variance at point $\mathbf{x}$ is obtained as:

$$\text{Var}[\hat{f}_H(\mathbf{x})] = \frac{\mathbf{e}^T \mathbf{e}}{n_H - (p + 1)} \mathbf{M}(\mathbf{x})^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{M}(\mathbf{x}) \quad (18)$$

where $\mathbf{M}(\mathbf{x}) = [f_L(\mathbf{x}) \xi_1(\mathbf{x}) \cdots \xi_p(\mathbf{x})]^T$ represents the design vector of linear regression at point $\mathbf{x}$.

Given these, the linear regression multi-fidelity surrogate model is easily applicable and can be implemented conveniently. Compared to the Gaussian process based multi-fidelity surrogate models, the training of the linear regression multi-fidelity surrogate model could be faster. In addition, this linear regression multi-fidelity surrogate model views the low-fidelity model as a basis, so it can also be easily applied to more than two low-fidelity models with different scale factors.

3. Numerical method and optimization algorithm

3.1. CFD method and validation

The RANS code used in this study employs Roe-FDS method [35] in space discretization and an implicit LU-SGS method [35] in time discretization. The two-equation $k-\omega$ SST turbulence model [36] is adopted for turbulence simulation.

In order to verify the reliability of the CFD code as the high-fidelity model, numerical simulations are conducted to the RAE 2822 airfoil model and the DLR-F6 wing-body model, then the computational results are compared with the experimental ones.

Fig. 5 shows the computational grid around RAE 2822 airfoil, grid number of the entire computational domain is about 150 thousand. Computational conditions are set as: $\alpha = 2.31^\circ$, $Ma = 0.729$, $Re = 2.13 \times 10^7$.

Fig. 6 shows the computational pressure coefficient distributions of RAE 2822 airfoil compared with experimental ones [37].

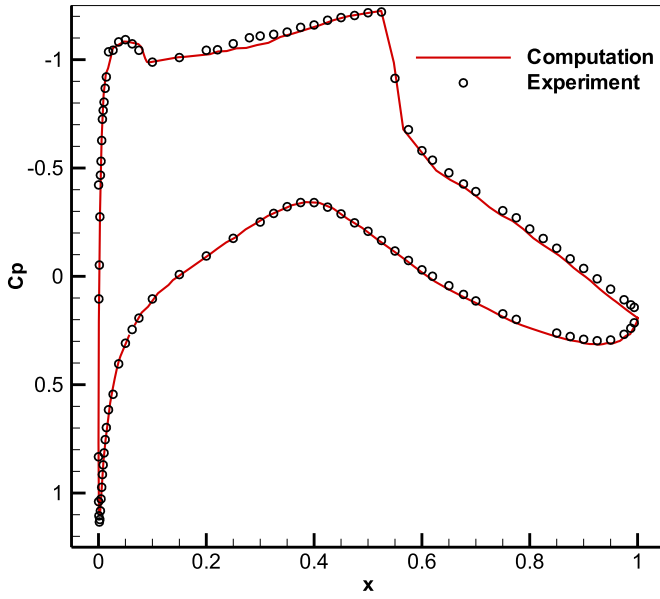


Fig. 6. Comparison of the pressure coefficient distributions for RAE 2822 airfoil.

Table 1

Comparison of the aerodynamic coefficients for RAE 2822 airfoil.

	C_l	C_d	C_m
Computation	0.7246	0.0119	−0.0884
Experiment	0.7310	0.0121	−0.0901
Error	0.88%	1.65%	1.89%

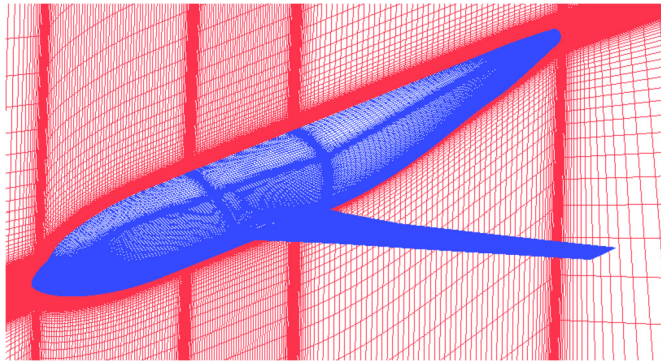


Fig. 7. Computational grid near DLR-F6 wing-body semi-model.

As can be seen, the computational results of pressure coefficients are in great agreement with the experimental ones.

Table 1 shows the comparison of aerodynamic coefficients between the computational results and experimental ones [38]. Errors of lift coefficient, drag coefficient and moment coefficient are only 0.88%, 1.65% and 1.89%, respectively. Thus, the computational results are greatly in accordance with experimental ones.

Fig. 7 shows the computational grid near DLR-F6 wing-body semi-model. Grid number of the entire computational domain is about 22 million, and the computational condition is set as: $Re = 3.0 \times 10^6$.

Table 2 shows the comparison of aerodynamic coefficients between the computational results and experimental ones [39]. As can be observed, the maximum error of C_l is only 0.28%, while the maximum error of C_d is only 1.02%. Hence, the computational results are in great agreement with the experimental ones.

From the above computational results of RAE 2822 airfoil and DLR-F6 wing-body compared with the experimental ones, it can

Table 2

Comparison of aerodynamic coefficients for DLR-F6 wing-body.

Case No.		C_l	C_d
Case 1	Computation	0.4988	0.0292
	Experiment	0.5000	0.0295
$\alpha = 0.52^\circ$	Error	0.24%	1.02%
Case 2	Computation	0.4992	0.0296
	Experiment	0.5000	0.0298
$\alpha = 0.47^\circ$	Error	0.16%	0.67%
Case 3	Computation	0.4986	0.0297
	Experiment	0.5000	0.0300
$\alpha = 0.41^\circ$	Error	0.28%	1.00%

be drawn that the CFD code used in this study is adequate for the high-fidelity model.

3.2. Optimization algorithm and validation

In this study, an improved PSO algorithm is employed as optimization method, and the multi-fidelity surrogate model is embedded to evaluate the fitness functions. In order to improve the convergence rate and global searching ability, some improvements are implemented to the standard PSO algorithm [40]: 1) the LHS method [41] is introduced to distribute original positions of the particles more dispersedly; 2) a second-order oscillating scheme [42] is used for updating velocities and positions of the particles; 3) the mutation operation [43] of GA is employed after positions of the particles are updated.

Fig. 8 shows the flowchart of the improved PSO algorithm. By employing a second-order oscillating scheme, the improved velocity and position updating equations are obtained as follows:

$$\begin{cases} v_j^{(i)}(t+1) = w(t)v_j^{(i)}(t) \\ \quad + c_1 r_1 [p_j^{(i)}(t) - (1 + \xi_1)x_j^{(i)}(t) + \xi_1 x_j^{(i)}(t-1)] \\ \quad + c_2 r_2 [p_j^{(g)}(t) - (1 + \xi_2)x_j^{(i)}(t) + \xi_2 x_j^{(i)}(t-1)], \\ \quad 1 \leq i \leq n, 1 \leq j \leq D \\ x_j^{(i)}(t+1) = x_j^{(i)}(t) + v_j^{(i)}(t+1) \end{cases} \quad (19)$$

where the subscript “j” denotes the j-th dimension of the particle, the superscript “(i)” denotes the i-th particle, and the superscript “(g)” denotes the particle with global optimum. c_1 , c_2 are positive constants called acceleration factors, r_1 , r_2 are random numbers in the domain (0, 1), w is the inertia weight factor, and ξ_1 , ξ_2 are random numbers. In order to make the improved PSO algorithm have a strong global searching ability and a fast convergence rate, ξ_1 and ξ_2 are assigned as the following rules:

$$\begin{cases} \xi_1 < \frac{2\sqrt{c_1 r_1} - 1}{c_1 r_1}, \quad \xi_2 < \frac{2\sqrt{c_2 r_2} - 1}{c_2 r_2} & \text{At the earlier stage} \\ \xi_1 \geq \frac{2\sqrt{c_1 r_1} - 1}{c_1 r_1}, \quad \xi_2 \geq \frac{2\sqrt{c_2 r_2} - 1}{c_2 r_2} & \text{At the later stage} \end{cases} \quad (20)$$

In order to balance global and local searching abilities, the inertia weight factor w is set to vary as follows:

$$w(t) = \frac{T-t}{T} (w_{\text{initial}} - w_{\text{final}}) + w_{\text{initial}} \quad (21)$$

where T is the maximum iterations, w_{initial} is the initial inertia weight factor, and w_{final} is the final inertia weight factor at the last iteration. In this study, w_{initial} and w_{final} are set to be 0.9 and 0.4, respectively.

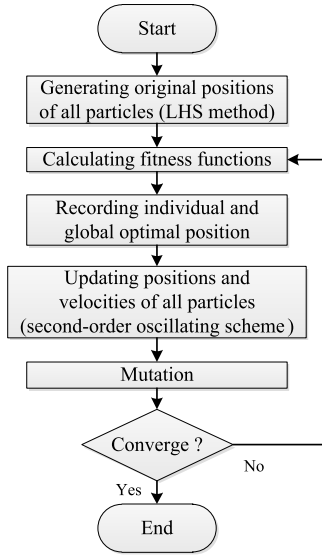


Fig. 8. Flowchart of the improved PSO algorithm.

For the purpose of overcoming the precocity problem that PSO algorithm falls into local optima as the swarm diversity drops too fast, the mutation operation of GA is introduced to enhance the global searching capability.

The mutation operation is implemented as follows:

- (1) Pick out the optimal particles in the swarm and select three particles for the mutation process randomly;
- (2) Generate a random number to determine the mutation point;
- (3) At the mutation point, carry out the mutation operation as follows:

$$x_j^{(i)} = x_j^{(rn1)} + \alpha(x_j^{(g)} - x_j^{(i)} + x_j^{(rn2)} - x_j^{(rn3)}) \quad (22)$$

where $x_j^{(rn1)}$, $x_j^{(rn2)}$ and $x_j^{(rn3)}$ represent the position of the random particles $rn1$, $rn2$ and $rn3$ in the j -th dimension, respectively. α is the differential scaling factor, and is set to be 0.01.

- (4) Repeat steps (1)–(3) until all particle mutations are completed.

By introducing the mutation operation, the diversity of the swarm is increased, and thus the algorithm is expected to obtain the global optimal solution and have strong robustness.

In order to validate the efficiency of the improved PSO algorithm, aerodynamic design optimizations for airfoil are performed via both the improved PSO algorithm and the standard PSO algorithm, and the results are compared.

The design conditions are set as: $Cl = 0.7$, $Ma = 0.71$, $Re = 2.3 \times 10^7$. The optimization problem is described as follows:

$$\begin{cases} \text{fitness} = Cd \\ \min \text{fitness} \\ \text{s.t.} \begin{cases} \alpha \leq 3^\circ \\ \text{thickness} = 13.02\% \end{cases} \end{cases} \quad (23)$$

The number of particles is 30, and CFD simulations are employed during the optimization process to evaluate the fitness functions. Fig. 9 shows the comparison of the fitness convergence history between the standard PSO algorithm and the improved PSO algorithm for the airfoil optimization. As can be seen, the optimization via the improved PSO algorithm converges 1 iteration earlier than that via the standard PSO algorithm, and also its final fitness value namely the drag coefficient is much lower than

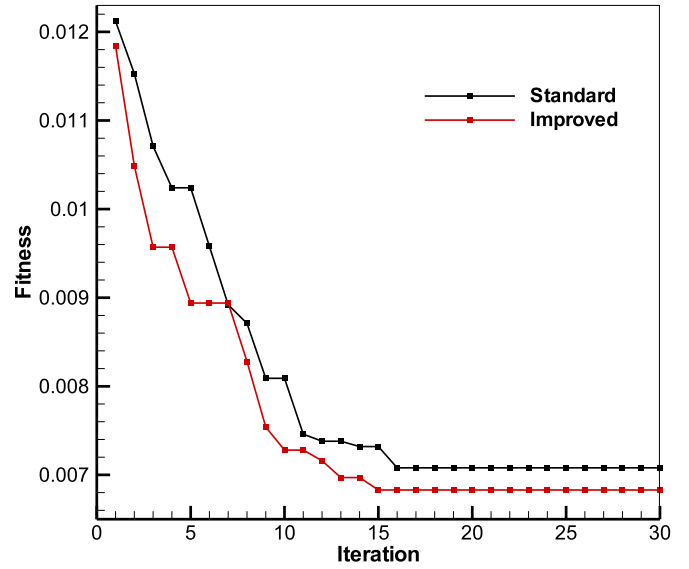


Fig. 9. Comparison of the fitness convergence history between the standard PSO algorithm and the improved algorithm for the airfoil optimization. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

that via the standard PSO algorithm. The comparison results indicate that the improved PSO algorithm has a faster convergence rate and a stronger global optimum searching capacity than the standard PSO algorithm.

4. Verification of the multi-fidelity surrogate model

In this section, the proposed multi-fidelity surrogate model is verified through the McCormick function, the dataset from fluidized-bed process in Ref. [44], and predictions of lift-to-drag ratios for a set of airfoils. Moreover, for predicting lift-to-drag ratios of a set of airfoils, BP network-based surrogate model, Kriging surrogate model, DBN-based surrogate model and DBN-based multi-fidelity surrogate model are employed, and the prediction results are compared among these surrogate models.

4.1. Case verification 1: McCormick function

The McCormick function is expressed as follows:

$$f_H(\mathbf{x}) = \sin(x_1 + x_2) + (x_1 - x_2)^2 - \frac{3}{2}x_1 + \frac{5}{2}x_2 + 1 \quad (24)$$

$$-1.5 \leq x_1 \leq 4; -3 \leq x_2 \leq 3$$

Here, the function itself is taken as the high-fidelity model. The low-fidelity model namely the DBN model is built with one input layer, 20 hidden layers and one output layer. The input layer has 2 units (namely $\mathbf{x} = (x_1, x_2)$), each hidden layer has 2 units, and the output layer has 1 unit (namely $f_L(\mathbf{x})$). In order to train the DBN, a dataset including 100 training samples is generated and denoted as $\mathbf{X}_L = \{\mathbf{x}_L^{(1)}, \dots, \mathbf{x}_L^{(100)}\}$, then high-fidelity results for the dataset are obtained through the high-fidelity model and denoted as $f_H(\mathbf{X}_L)$. After the DBN is well trained, another dataset including 20 samples is generated and denoted as $\mathbf{X}_H = \{\mathbf{x}_H^{(1)}, \dots, \mathbf{x}_H^{(20)}\}$, and the high-fidelity results are obtained via the high-fidelity model and denoted as $f_H(\mathbf{X}_H)$.

Given the above, the multi-fidelity surrogate model is established consequently. In order to evaluate the effect of the multi-fidelity surrogate model on predicting function values, a test dataset consisting of 10 samples is generated and predicted by the multi-fidelity surrogate model. Fig. 10 shows the comparison

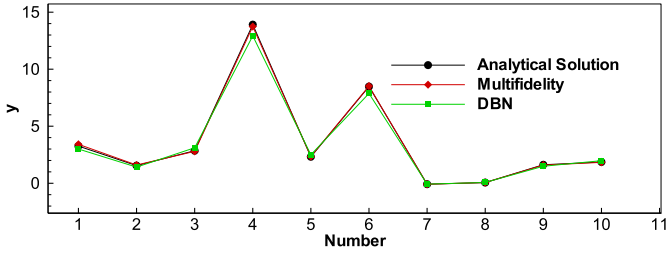


Fig. 10. Comparison of McCormick function among analytical values, multi-fidelity surrogate predictions and DBN predictions.

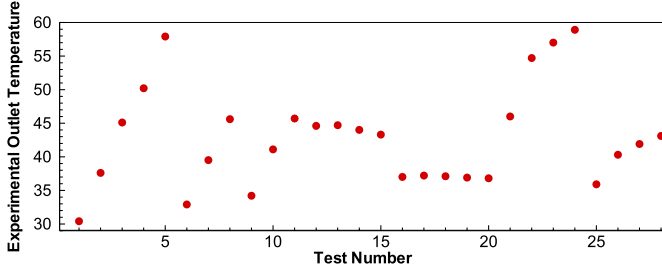


Fig. 11. Experimental values [44] of outlet air temperature from fluidized-bed process.

among the analytical values, the multi-fidelity surrogate predictions and the DBN (low-fidelity model) predictions for the test dataset. As can be seen, the multi-fidelity surrogate model obtains more accurate predictions than DBN model, and the multi-fidelity surrogate predictions are greatly in accordance with the analytical values.

4.2. Case verification 2: dataset from fluidized-bed process

The dataset from fluidized-bed process in Ref. [44] is used for verification of the multi-fidelity surrogate model. In the literature, the quantity of interest is the temperature of the steady-state thermodynamic operation point (namely the outlet air temperature T_2) for a fluidized-bed process. Six variables, which affect the outlet air temperature, are investigated. The six variables are air relative humidity (φ_r), room temperature (T_r), inlet air temperature (T_1), flow rate of the coating solution (R_{sol}), pressure of atomization air (P_{at}), and fluid velocity of the fluidization air (V_{fl}), respectively.

Experiments were conducted under 28 different process conditions with coating solution used for distilled water at room temperature. Fig. 11 shows the 28 experimental results [44] of the outlet air temperature from the fluidized-bed process.

In this study, the 28 experimental samples are divided into three portions as shown in Table 3. The low-fidelity model namely the DBN model is built with one input layer, 16 hidden layers and one output layer. The input layer has 6 units (namely $\mathbf{x} = (\varphi_r, T_r, T_1, R_{sol}, P_{at}, V_{fl})$), each hidden layer has 6 units, and the output layer has 1 unit (namely $T_2 = f_L(\mathbf{x})$). The 19 samples shown in Table 3 are used to train the DBN and denoted as $\mathbf{X}_L = \{\mathbf{x}_L^{(1)}, \dots, \mathbf{x}_L^{(19)}\}$, and the corresponding experimental values of the outlet air temperature are denoted as $f_H(\mathbf{X}_L)$. After the DBN is well trained, the 5 samples shown in Table 3 are used for constructing the multi-fidelity surrogate model and denoted as $\mathbf{X}_H = \{\mathbf{x}_H^{(1)}, \dots, \mathbf{x}_H^{(5)}\}$, and the corresponding experimental values of the outlet air temperature are denoted as $f_H(\mathbf{X}_H)$.

Fig. 12 shows the comparison of T_2 among the experimental results [44], the multi-fidelity surrogate predictions and the DBN (low-fidelity model) predictions for the test group. As can be seen, the multi-fidelity surrogate predictions are closer to the experimental results than the DBN model, and match very well with the experimental results.

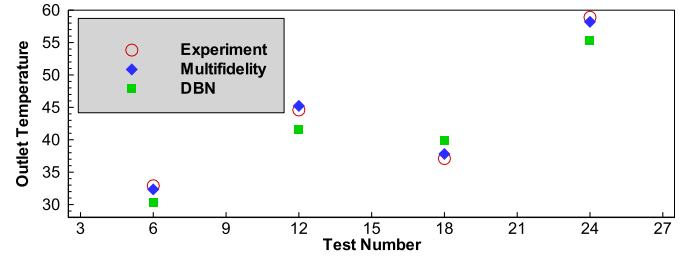


Fig. 12. Comparison of T_2 among experimental results [44], multi-fidelity surrogate predictions and DBN predictions.

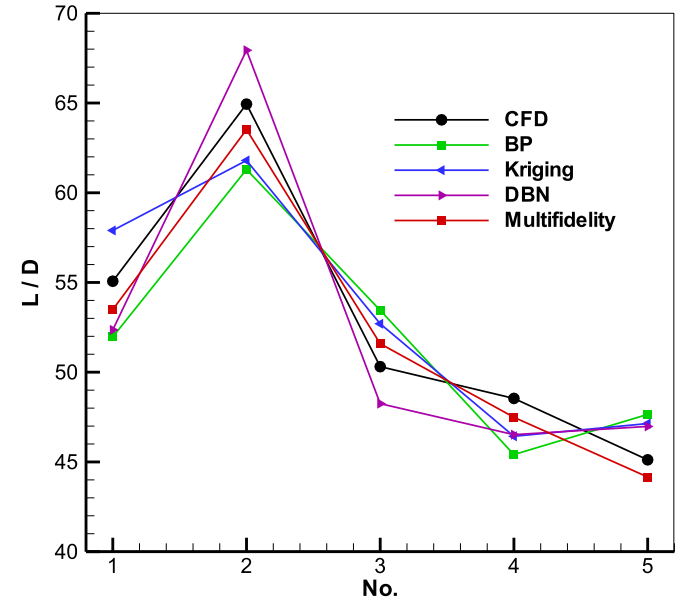


Fig. 13. Prediction results of lift-to-drag ratios via different surrogate models compared with CFD results.

4.3. Case verification 3: predictions of lift-to-drag ratios for airfoils

Firstly, a database containing 30 airfoils is established via LHS method for training BP network-based surrogate model, Kriging surrogate model and DBN-based surrogate model. All the airfoils are parameterized with the six-order CST method [35], and thus the number of geometric parameters for each airfoil is 14. Lift-to-drag ratio of each airfoil is obtained through CFD simulations, and the computational conditions are set as: $\alpha = 0^\circ$, $Ma = 0.7$, $Re = 3 \times 10^6$. The geometric parameters of all the airfoils in the database are denoted as $\mathbf{X}_L = \{\mathbf{x}_L^{(1)}, \dots, \mathbf{x}_L^{(30)}\}$, and lift-to-drag ratios of these airfoils via CFD simulations are denoted as $f_H(\mathbf{X}_L)$. The DBN model is built with one input layer, 20 hidden layers and one output layer. The input layer has 14 units (namely the 14 geometric parameters), each hidden layer has 14 units, and the output layer has 1 unit (namely lift-to-drag ratio). Then, a test group containing 5 airfoils is generated, and CFD simulations are performed to the 5 airfoils under the same computational conditions. In order to construct the multi-fidelity surrogate model, an independent group containing 8 airfoils is generated. The geometric parameters of the 8 airfoils are denoted as $\mathbf{X}_H = \{\mathbf{x}_H^{(1)}, \dots, \mathbf{x}_H^{(8)}\}$, and lift-to-drag ratios of them are obtained via CFD simulations and denoted as $f_H(\mathbf{X}_H)$. Finally, predictions for lift-to-drag ratios are conducted to the 5 airfoils in the test group via different surrogate models.

Fig. 13 shows the prediction results of lift-to-drag ratios for the 5 airfoils via different surrogate models compared with CFD results. As can be observed, the predictions by the multi-fidelity surrogate model are closest to the results by CFD simulations.

Table 3

Three portions of the 28 experimental samples.

Test No.	Purpose
3, 9, 15, 21, 27 (5 in total)	Used as high-fidelity data to construct the multi-fidelity surrogate model
1, 2, 4, 5, 7, 8, 10, 11, 13, 14, 16, 17, 19, 20, 22, 23, 25, 26, 28 (19 in total)	Used as training samples for the low-fidelity model
6, 12, 18, 24 (4 in total)	Used as test group

Table 4

Average errors between the predictions by different surrogate models and CFD results.

Surrogate model	BP	Kriging	DBN	Multi-fidelity
Average error	5.91%	4.71%	4.38%	2.39%

Table 4 shows the average errors between predictions by different surrogate models and CFD results. Average error of the multi-fidelity surrogate model is the smallest, which is only 2.39%. BP network-based surrogate model has the largest average error of 5.91%. Average error of DBN-based surrogate model is slightly smaller than that of Kriging surrogate model. Therefore, from the comparison results of average errors, the multi-fidelity surrogate model proposed in this study obtains more accurate predictions than the other three surrogate models. Such results demonstrate that the multi-fidelity surrogate model is highly reliable, and it lays the foundation of aerodynamic design optimizations.

Therefore, from the above three verification cases, the proposed multi-fidelity surrogate model is proved to be an excellent prediction model. Thus, it can be applied to the aerodynamic design optimizations for airfoil and wing, which are discussed in section 5.

5. Application of the multi-fidelity surrogate model to airfoil and wing design optimizations

In this section, the proposed multi-fidelity surrogate model is applied to the aerodynamic robust optimizations for airfoil and wing under uncertainty of Mach number. A RANS CFD code is employed as the high-fidelity model, while two DBN models are built as low-fidelity models for airfoil and wing optimizations, respectively. The five-order CST (class/shape function transformation) method [45] is used for airfoil parameterization, and an improved PSO algorithm is adopted as optimization method for both airfoil and wing optimizations.

5.1. Aerodynamic design optimization for airfoil

The airfoil is designed for the wing of a high-subsonic commercial aircraft. The cruise Mach number of the aircraft is 0.785, while the drag divergence Mach number is 0.805. In order to improve aerodynamic performance of the airfoil, robust optimization is conducted in the Mach number range near the cruise Mach number up to the drag divergence Mach number.

Since cruise conditions of the wing are set as: $Ma = 0.785$, $Re = 2.3 \times 10^7$, $Cl_{wing} = 0.515$, according to rule of cosine [46], cruise conditions of the airfoil are obtained as: $Ma = 0.71$, $Re = 2.3 \times 10^7$, $Cl = 0.7$. The airfoil is parameterized with the five-order CST method, and thus the number of design variables is 12 corre-

spondingly. Table 5 shows the design space of the five-order CST variables for the robust optimization of the airfoil.

According to rule of cosine, the Mach number range for the airfoil robust optimization is set to be [0.69, 0.73], which corresponds to the Mach number range of [0.765, 0.805] for the wing design.

The design conditions are set as: $Cl = 0.7$, $Re = 2.3 \times 10^7$. Then the robust optimization for the airfoil via the improved PSO algorithm is expressed as follows:

$$\begin{cases} \text{fitness} = \mu_{Cd}^2 + \sigma_{Cd}^2 \\ \text{min fitness} \\ \text{s.t.} \begin{cases} \alpha \leq 3^\circ \\ \text{thickness} = 12.5\% \end{cases} \end{cases} \quad (25)$$

$$\mu_{Cd} = \frac{1}{5}(Cd_{Ma=0.69} + Cd_{Ma=0.70} + \dots + Cd_{Ma=0.73})$$

$$\sigma_{Cd} = \sqrt{\frac{1}{4}[(Cd_{Ma=0.69} - \mu_{Cd})^2 + (Cd_{Ma=0.70} - \mu_{Cd})^2 + \dots + (Cd_{Ma=0.73} - \mu_{Cd})^2]} \quad (26)$$

where μ_{Cd} is the expectation of C_d in the Mach number range of [0.69, 0.73], and σ_{Cd} is the standard deviation of C_d in the Mach number range of [0.69, 0.73].

Fig. 14 shows the flowchart of the robust optimization for the airfoil and wing. For airfoil optimization, the number of particles is set to be 50. Firstly, 150 airfoils are generated as the original samples via LHS method and CST method, and then parallel computations are conducted to these original samples via the RANS code under design conditions. Next, 80% of the original samples (namely 120 airfoils) are used for building and training the DBN, and the remaining 20% of the original samples (namely 30 airfoils) are used as high-fidelity data for construction of the multi-fidelity surrogate model. Finally, the multi-fidelity surrogate model is embedded into the improved PSO algorithm framework to evaluate fitness functions. Before the optimization is converged, in each iteration, 20% of the particles are selected to update the multi-fidelity surrogate model by adopting the multiple-point infilling criteria, which selects 10% particles near the current optimal particles and 10% particles with the largest prediction variances.

Fig. 15 shows the structure of the DBN for the airfoil robust optimization. The DBN model is built with one input layer, 12 hidden layers and one output layer. The input layer has 12 units (namely the 12 CST parameters), each hidden layer has 12 units, and the output layer has 5 units (namely drag coefficients under five different Mach numbers).

Fig. 16 shows the fitness of the airfoil robust optimization varying with iterations. As can be observed, the optimization converges at iteration No. 17.

Assuming there is a CFD-based optimization which also converges at iteration No. 17 for the airfoil optimization, Table 6

Table 5

Ranges of the five-order CST variables.

Upper surface parameter No.	1	2	3	4	5	6
Range	[0.05, 0.2]	[0.05, 0.25]	[0.1, 0.3]	[0.1, 0.3]	[0.1, 0.3]	[0.1, 0.3]
Lower surface parameter No.	1	2	3	4	5	6
Range	[-0.2, -0.05]	[-0.25, 0.05]	[-0.3, -0.1]	[-0.4, -0.1]	[-0.3, -0.05]	[0.05, 0.2]

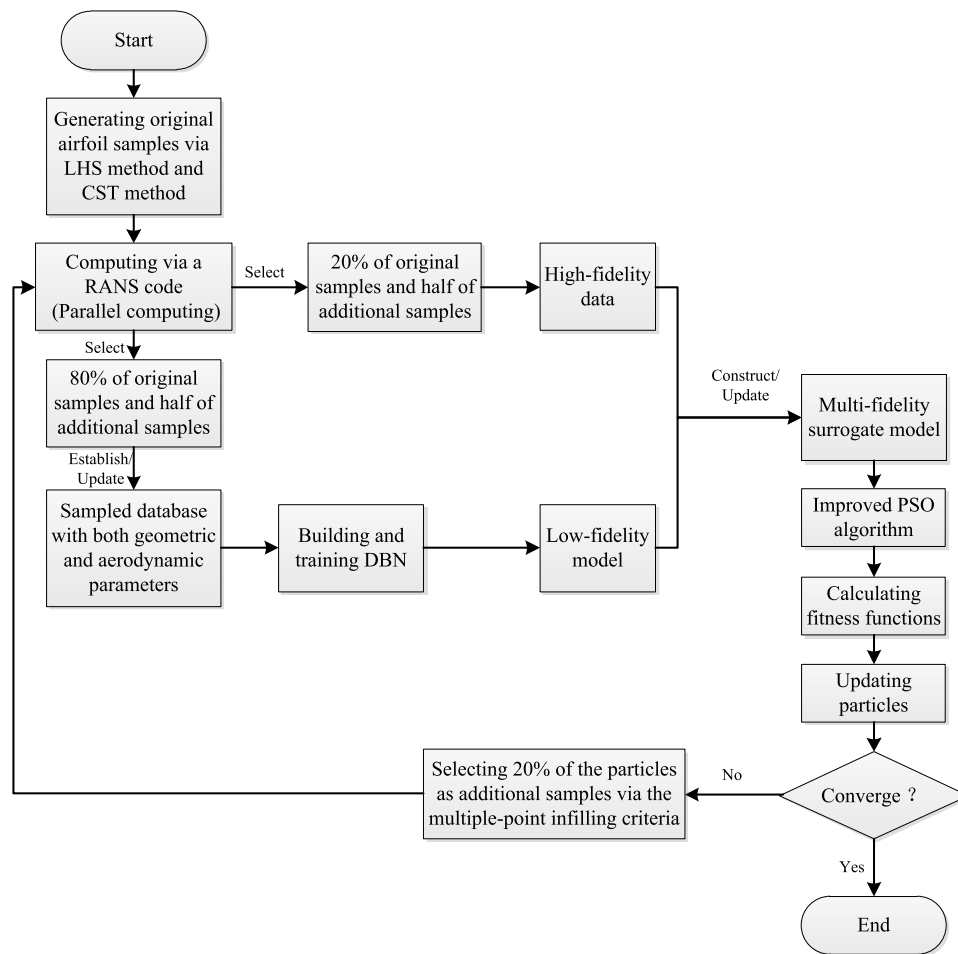


Fig. 14. Flowchart of the airfoil/wing design optimization.

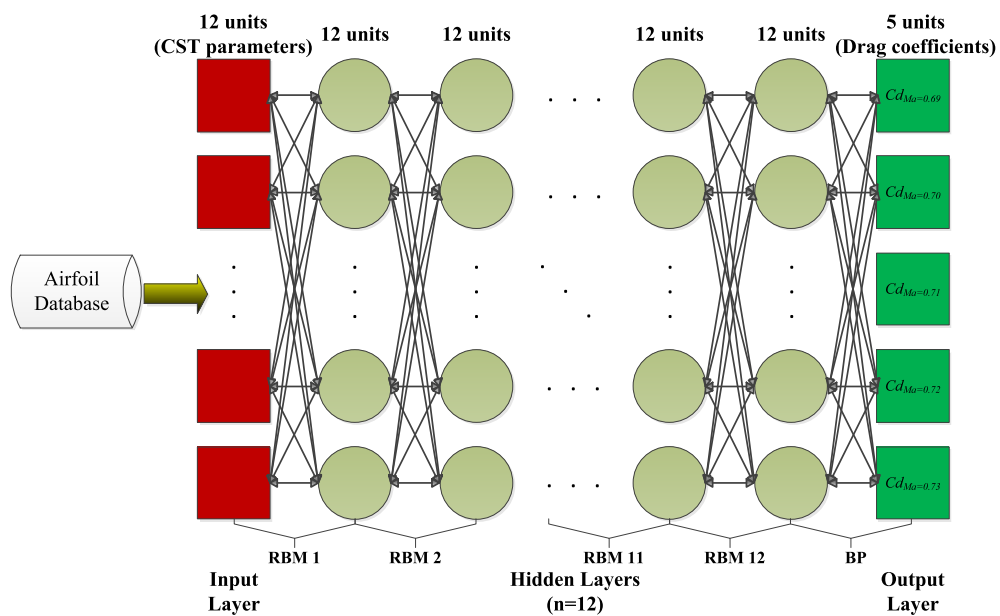


Fig. 15. Structure of the DBN for the airfoil design optimization.

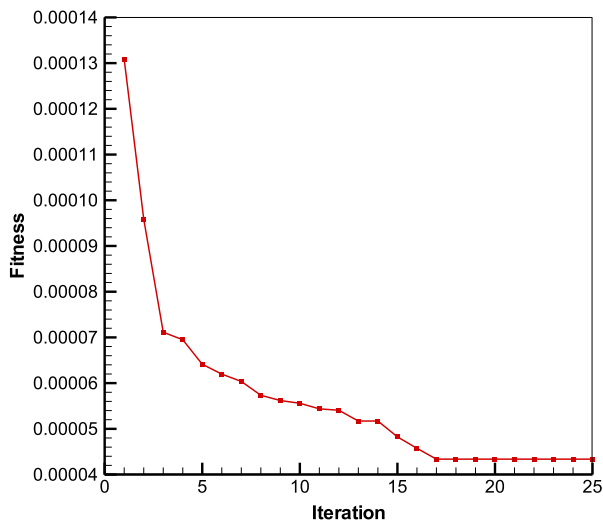


Fig. 16. Fitness of the airfoil design optimization varying with iterations.

shows the numbers of computations required for both the two airfoil optimizations until the optimization converges. As can be seen, the number of computations required for CFD-based optimization is 4250, while that required for multi-fidelity surrogate-based optimization is only 1600. Since the time of training the surrogate model is ignorable compared with CFD computation, the computing time is saved by about 62% through employing the multi-fidelity surrogate model.

During the optimization process, in order to investigate the accuracy of the multi-fidelity surrogate model, the global optimal particle in each iteration is picked out, and drag coefficients are compared between the multi-fidelity surrogate predictions and CFD results. Fig. 17 shows the comparisons of C_d between the multi-fidelity surrogate predictions and CFD results varying with iterations. As can be seen, at the earlier stage of the optimization, there are relatively big differences between the multi-fidelity surrogate predictions and CFD results for C_d under the five different Mach numbers, but the differences become smaller as the iteration number increases. At the later stage, the multi-fidelity surrogate predictions get very close to the CFD results. It can be drawn that,

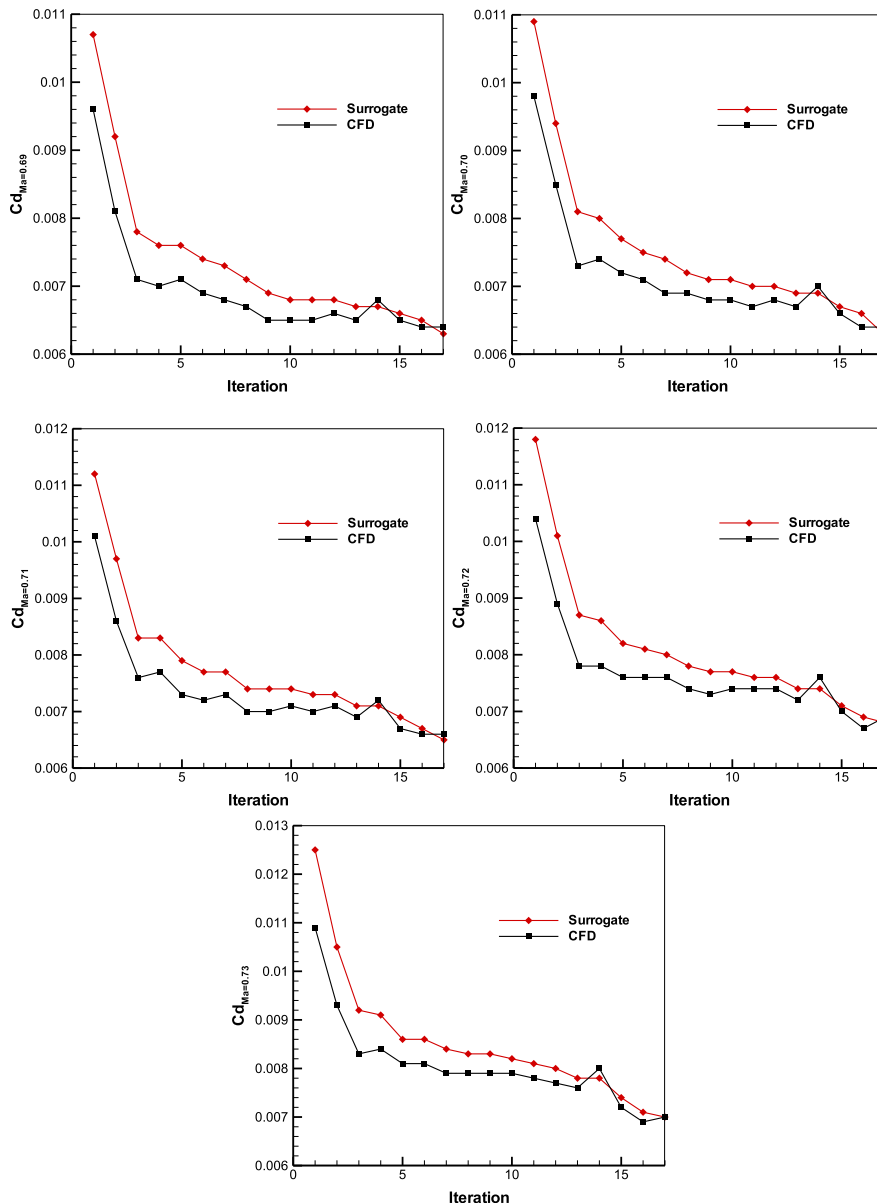


Fig. 17. Comparisons of C_d of the global optimal particles for the airfoil robust optimization.

Table 6

Numbers of computations required for both the two airfoil optimizations.

	Number of computations for training	Number of computations for each particle in each iteration	Number of computations for all particles in each iteration	Number of computations until optimization convergence
CFD-based optimization	/	5	250	4250
Multi-fidelity surrogate-based optimization	750	/	50	1600

the multi-fidelity surrogate model is updated and corrected as the iteration number increases, thus shows such close predictions to the CFD results.

Fig. 18 shows the geometric shapes of the initial airfoil and the optimized one. Fig. 19 shows the comparisons of C_d between the surrogate predictions and CFD results in the Mach number range of [0.69, 0.73] for the optimized airfoil. As can be seen, drag coefficients predicted by the multi-fidelity surrogate model is basically in accordance with those obtained by CFD simulations, and the variation tendency of the surrogate predictions is also in agreement with that of the CFD results. Therefore, the multi-fidelity surrogate model performs very well as a prediction model during the optimization process of the airfoil robust optimization, which is because the multi-fidelity surrogate model is updated and corrected in each iteration.

5.2. Aerodynamic design optimization for wing

As shown in Fig. 20, the wing is installed on a wing-body model, and geometric shape of the body remains unchanged during the wing optimization process. Semi span of the wing is 17.204 m, leading sweepback angle is 27.4217° , root chord is 6.18 m, and tip chord is 1.518 m. There are three control profiles positioned at the root, the kink and the tip, respectively. Each control profile is parameterized with the five-order CST method, and ranges of the five-order CST variables are given in Table 5. Besides the CST variables of the three control profiles, twist angles of the three control profiles are also deemed as design variables. Thus, the number of design variables reaches 39.

The Mach number range for the wing robust optimization is set to be [0.765, 0.805], and the design conditions are set as: $CL_{wingbody} = 0.575$, $Re = 2.3 \times 10^7$. Then the robust optimization for the wing via the improved PSO algorithm is expressed as follows:

$$\begin{cases} \text{fitness} = \mu_{CD}^2 + \sigma_{CD}^2 \\ \min \text{fitness} \\ \text{s.t.} \begin{cases} \alpha \leq 3^\circ \\ \text{thickness_root} = 15\% \\ \text{thickness_kink} = 12\% \\ \text{thickness_tip} = 10\% \end{cases} \end{cases} \quad (27)$$

$$\mu_{CD} = \frac{1}{5}(CD_{Ma=0.765} + CD_{Ma=0.775} + \dots + CD_{Ma=0.805})$$

$$\sigma_{CD} = \sqrt{\frac{1}{4}[(CD_{Ma=0.765} - \mu_{CD})^2 + (CD_{Ma=0.775} - \mu_{CD})^2 + \dots + (CD_{Ma=0.805} - \mu_{CD})^2]} \quad (28)$$

where μ_{CD} is the expectation of the drag coefficient in the Mach number range of [0.765, 0.805], and σ_{CD} is the standard deviation of the drag coefficient in the Mach number range of [0.765, 0.805].

The robust optimization for the wing follows the flowchart shown in Fig. 12. The optimized airfoil in section 4.3 is used at the three control profiles with thicknesses corrected to construct an initial configuration, and this initial configuration is treated as an initial particle for the improved PSO algorithm. The number of

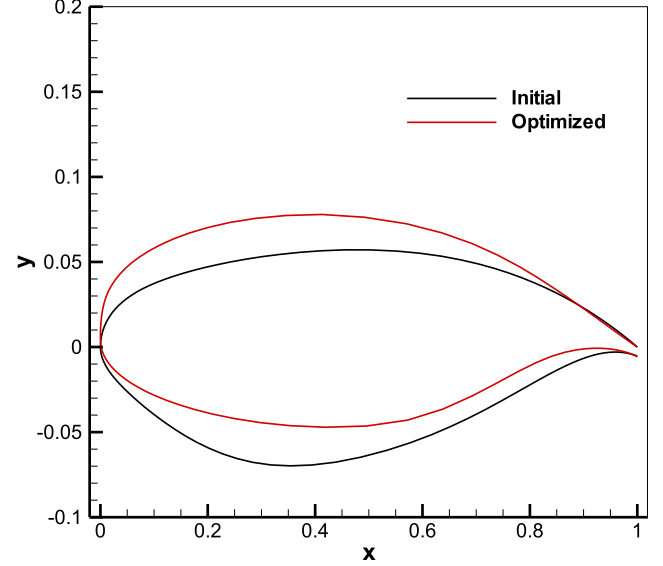


Fig. 18. Geometric shape comparison between the initial airfoil and the optimized one.

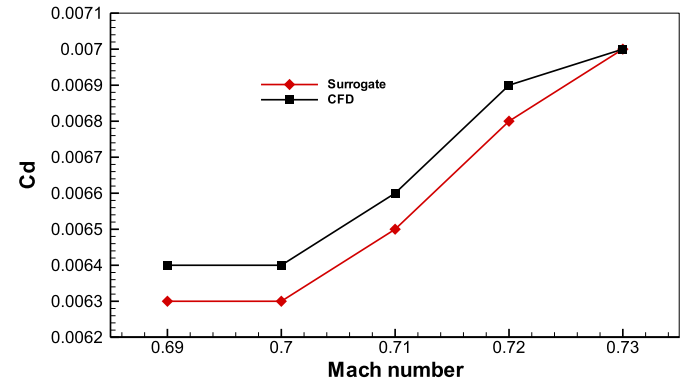


Fig. 19. Comparisons of C_d of the optimized airfoil between the surrogate predictions and CFD results in the Mach number range of [0.69, 0.73].

all particles is set to be 30. Firstly, 60 configurations are generated as the original samples via LHS method and CST method, and then parallel computations are performed to these original samples via the RANS code under design conditions. Next, 80% of the original samples (namely 48 configurations) are used for building and training the DBN, and the remaining 20% of the original samples (namely 12 configurations) are used as high-fidelity data for construction of the multi-fidelity surrogate model. Finally, the multi-fidelity surrogate model is embedded into the improved PSO algorithm framework to evaluate fitness functions. Similar to the airfoil optimization case, 20% of the particles are selected to update the multi-fidelity surrogate model by adopting the multiple-point infilling criteria in each iteration before the optimization is converged.

Fig. 21 shows the structure of the DBN for the wing robust optimization. The DBN model is built with one input layer, 20 hidden layers and one output layer. The input layer has 39 units (namely

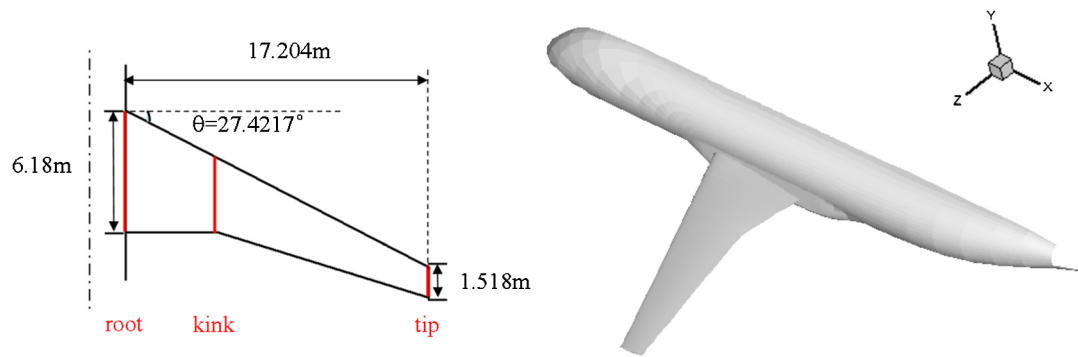


Fig. 20. Platform of the wing and schematic of the wing-body model.

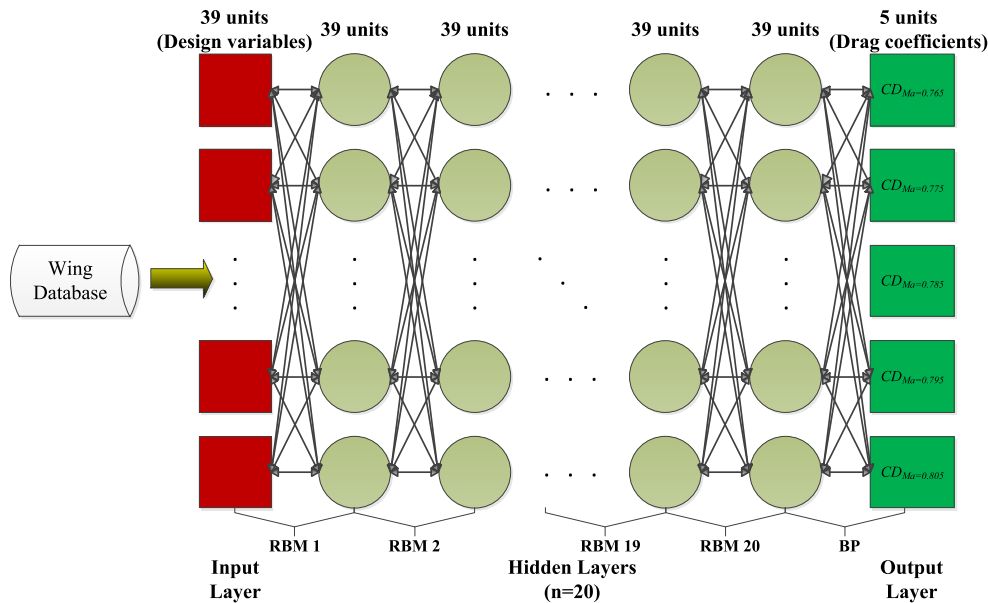


Fig. 21. Structure of the DBN for the wing design optimization.

the 39 design variables), each hidden layer has 39 units, and the output layer has 5 units (namely drag coefficients under five different Mach numbers).

Fig. 22 shows the fitness of the wing robust optimization varying with iterations. As can be seen, the optimization converges at iteration No. 15.

Assuming there is a CFD-based optimization which also converges at iteration No. 15 for the wing optimization, Table 7 shows the numbers of computations required for both the two wing optimizations until the optimization converges. As can be seen, the number of computations required for CFD-based optimization is 2250, while that required for multi-fidelity surrogate-based optimization is only 750. Since the time of training the surrogate model is ignorable compared with CFD computation, the computing time is saved by about 66% through employing the multi-fidelity surrogate model.

Similar to the airfoil robust optimization, in order to investigate the accuracy of the multi-fidelity surrogate model, the global optimal particle in each iteration is picked out during the optimization process, and drag coefficients are compared between the multi-fidelity surrogate predictions and CFD results. Fig. 23 shows the comparisons of C_D between the multi-fidelity surrogate predictions and CFD results varying with iterations. As can be observed, at the earlier stage of the optimization, there are still considerable differences between the multi-fidelity surrogate predictions and CFD results for C_D under the five different Mach numbers,

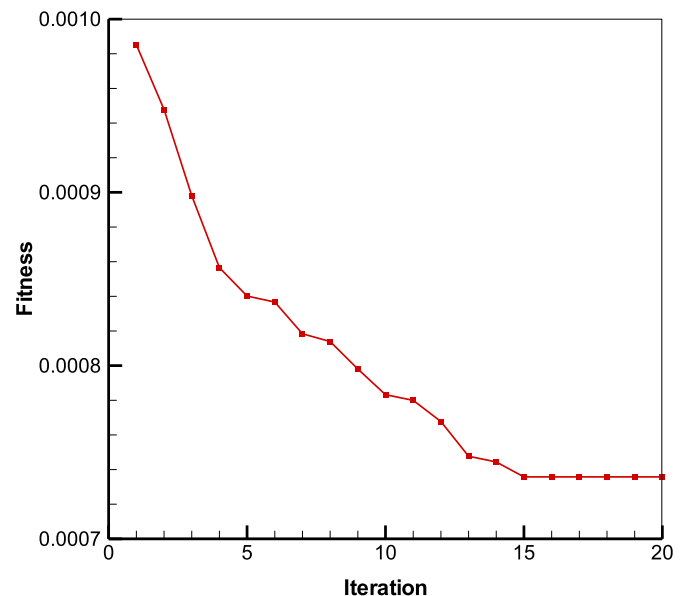


Fig. 22. Fitness of the wing design optimization varying with iterations.

but the differences tend to be smaller as the iteration number increases. At the later stage, the multi-fidelity surrogate predictions

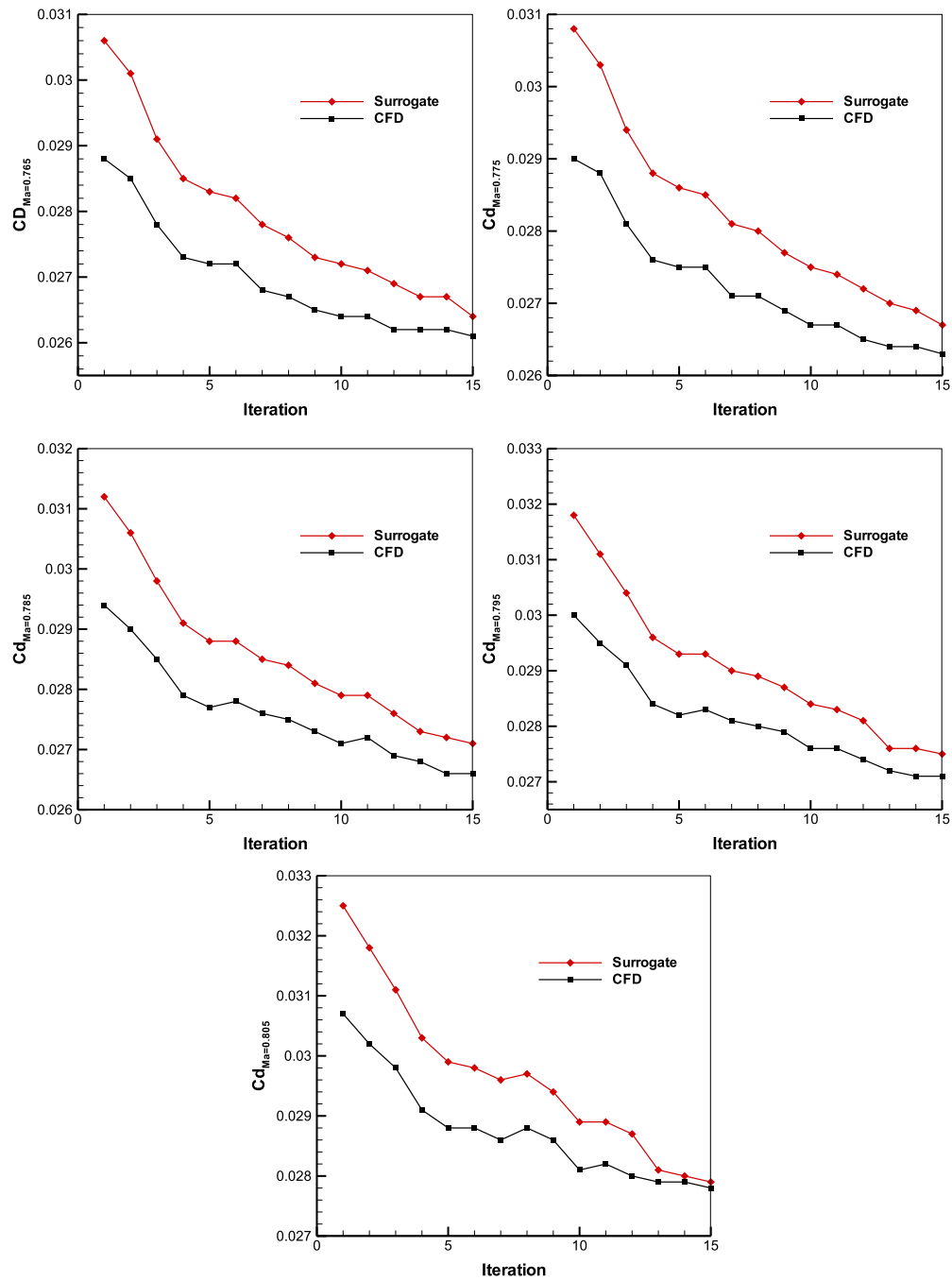


Fig. 23. Comparisons of C_D of the global optimal particles for the wing robust optimization.

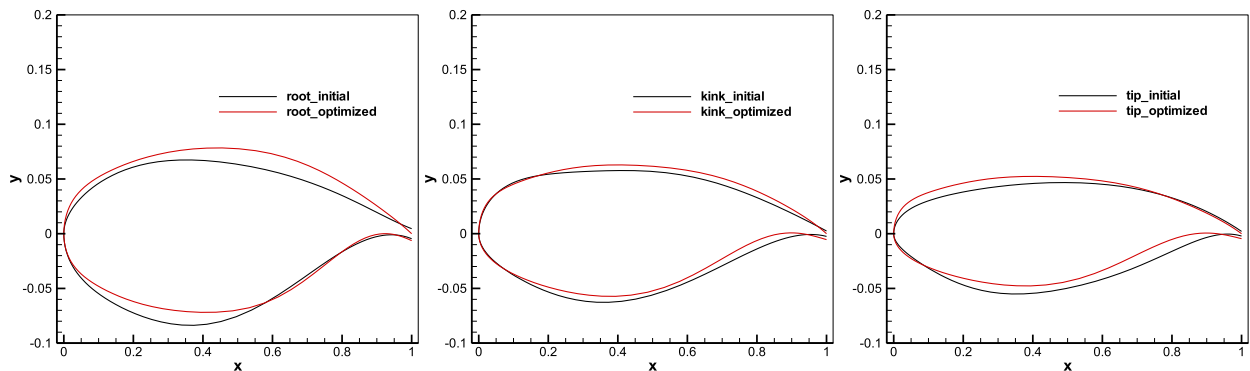
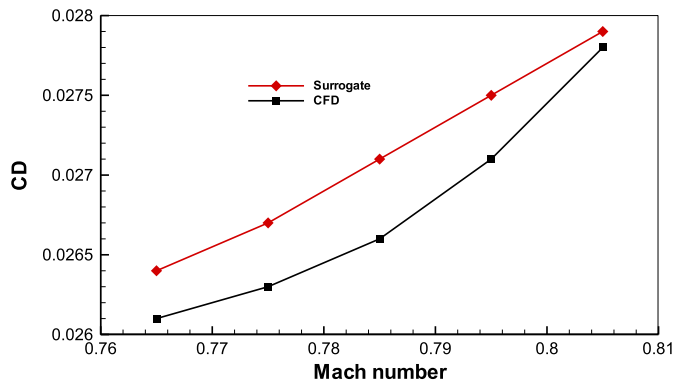


Fig. 24. Geometric shapes of the three initial and optimized control profiles.

Table 7

Numbers of computations required for both the two wing optimizations.

	Number of computations for training	Number of computations for each particle in each iteration	Number of computations for all particles in each iteration	Number of computations until optimization convergence
CFD-based optimization	/	5	150	2250
Multi-fidelity surrogate-based optimization	300	/	30	750

**Fig. 25.** Comparisons of C_D of the optimized wing between the surrogate predictions and CFD results in the Mach number range of [0.765, 0.805].

become close to the CFD results. It can be inferred that, such close predictions to the CFD results at the later stage result from that the multi-fidelity surrogate model is updated and corrected continuously in the optimization process.

Fig. 24 shows the geometric shapes of the three initial and optimized control profiles. Fig. 25 shows the comparisons of C_D between the surrogate predictions and CFD results in the Mach number range of [0.765, 0.805] for the optimized wing. As can be observed, drag coefficients predicted by the multi-fidelity surrogate model match well with those obtained by CFD simulations. In the meantime, the variation tendency of the surrogate prediction is also in accordance with that of the CFD result. From the above, because the multi-fidelity surrogate model is updated and corrected continuously during the optimization process, it serves very well as a prediction model for the wing robust optimization.

6. Conclusion

In this study, the robust optimizations under uncertainty of Mach number for airfoil and wing are performed via a multi-fidelity surrogate-based optimization framework.

A linear regression multi-fidelity surrogate model is built by employing DBN as the low-fidelity model. The DBN model is trained by utilizing the k -step contrastive divergence algorithm. The well trained DBN model along with the data obtained by high-fidelity model are combined to construct the final multi-fidelity surrogate model.

In order to verify the effect of the proposed multi-fidelity surrogate model, the McCormick function, the dataset from fluidized-bed process and predictions of lift-to-drag ratios for a set of airfoils are selected as verification cases. Verification results of the three cases indicate that, the proposed multi-fidelity surrogate model is highly reliable as a prediction model.

The multi-fidelity surrogate model is embedded into an improved PSO algorithm to evaluate fitness functions and applied to the robust optimizations under uncertainty of Mach number for airfoil and wing. The improvements to the standard PSO algorithm include introducing the LHS method to distribute original positions of the particles, using a second-order oscillating scheme to update velocities and positions of the particles, and implementing the mutation operation after positions of the particles are updated.

During the process of the multi-fidelity surrogate-based optimization, the multiple-point infilling criteria is adopted to update the multi-fidelity surrogate model. Precisely because the multi-fidelity surrogate model is updated and corrected continuously, the multi-fidelity surrogate predictions tend to approach the CFD results as the iteration number increases. Optimization results of airfoil and wing indicate that the multi-fidelity surrogate model performs very well as a prediction model, and thus improves the optimization efficiency significantly.

Declaration of Competing Interest

The authors declared that they have no conflicts of interest to this work.

Acknowledgements

The authors thank the reviewers for their critical and constructive review of the manuscript.

The authors are grateful for the support by Aeronautical Science Foundation of China (No. 2015ZBP9002).

This article is funded by China Scholarship Council (No. 201706105033).

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